

Long Term Trends in Melanoma Mortality in New Zealand

Before and after the introduction of modern systemic therapies

Population time series analysis of age standardised mortality from 1954 to 2023

Report type

Independent portfolio analysis

Prepared by

Zohreh Riahi

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Data source

Health New Zealand Cancer Web Tool

Executive Summary

The central finding is that melanoma mortality in New Zealand fell sharply around the 2016 treatment transition and remained lower through 2023. The pattern is visible in the national data, persists under several lag assumptions, and appears in both men and women.

The analysis identifies a strong temporal association. It does not prove that publicly funded systemic therapies caused the entire reduction because the dataset contains no patient level information on treatment exposure, stage, or survival.

Measure	Result	Meaning
Recent mortality	32.3% lower	Mean all sex rate in 2021 to 2023 versus 2012 to 2015
Estimated 2017 level change	-1.523 per 100,000	95% CI -1.765 to -1.281 $p < 0.001$
Post introduction slope	-0.046 per year	Mortality continued to decline after the initial reduction $p = 0.001$
Model fit	R squared 0.913	The primary model explained 91.3% of annual variation

Answer to the Research Question

The available evidence supports a sustained rather than temporary reduction. The observed rates remained about 3.2 to 3.7 deaths per 100,000 during 2017 to 2023, compared with a mean of 4.97 during 2012 to 2015.

Implications for Treatment Innovation

The population data show a measurable shift in melanoma outcomes after effective systemic therapies became publicly funded. The result is relevant to evidence based discussion of treatment innovation and population outcomes. Product specific attribution would require patient level treatment and survival data.

Study Context and Analytical Approach

Research Question

Was the introduction of modern systemic therapies for melanoma in New Zealand during 2016 associated with a temporary or sustained change in melanoma mortality

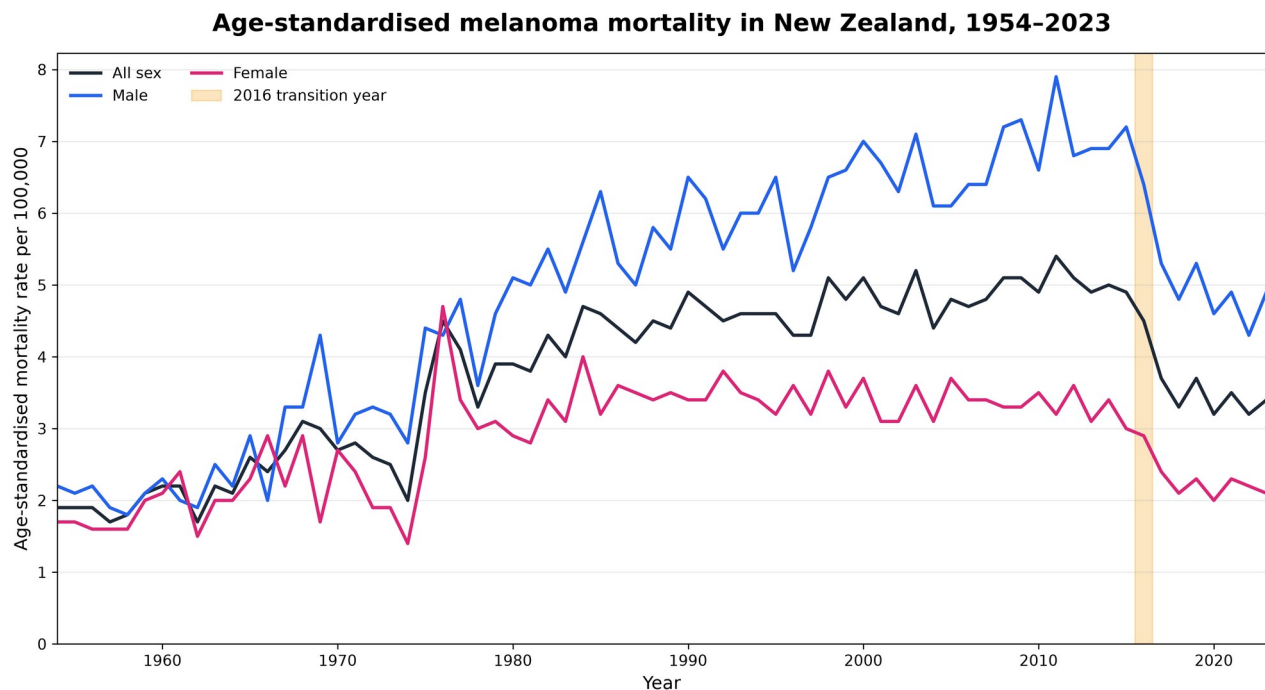
Data and Treatment Context

The Health New Zealand Cancer Web Tool supplied annual melanoma deaths and directly age standardised mortality rates for 1954 to 2023. The rates use the WHO standard world population and are reported for all sexes, men, and women. The source classifies melanoma as ICD 10 C43. Data for 2022 are provisional and data for 2023 are preliminary.

Nivolumab and pembrolizumab became publicly funded for eligible patients with advanced melanoma during 2016. The analysis therefore treated 2016 as a transition year, defined years through 2015 as pre introduction, and began the main post introduction period in 2017.

Methods

- Descriptive trends used the full 1954 to 2023 series.
- The primary segmented regression used all sex rates from 2000 to 2023 and excluded 2016.
- The model estimated the pre introduction slope, the level change in 2017, and the change in slope after 2017.
- Heteroskedasticity and autocorrelation consistent standard errors used two lags.
- Sensitivity models began the post period in 2017, 2018, or 2019.
- Separate models assessed whether the direction of change was consistent in men and women.

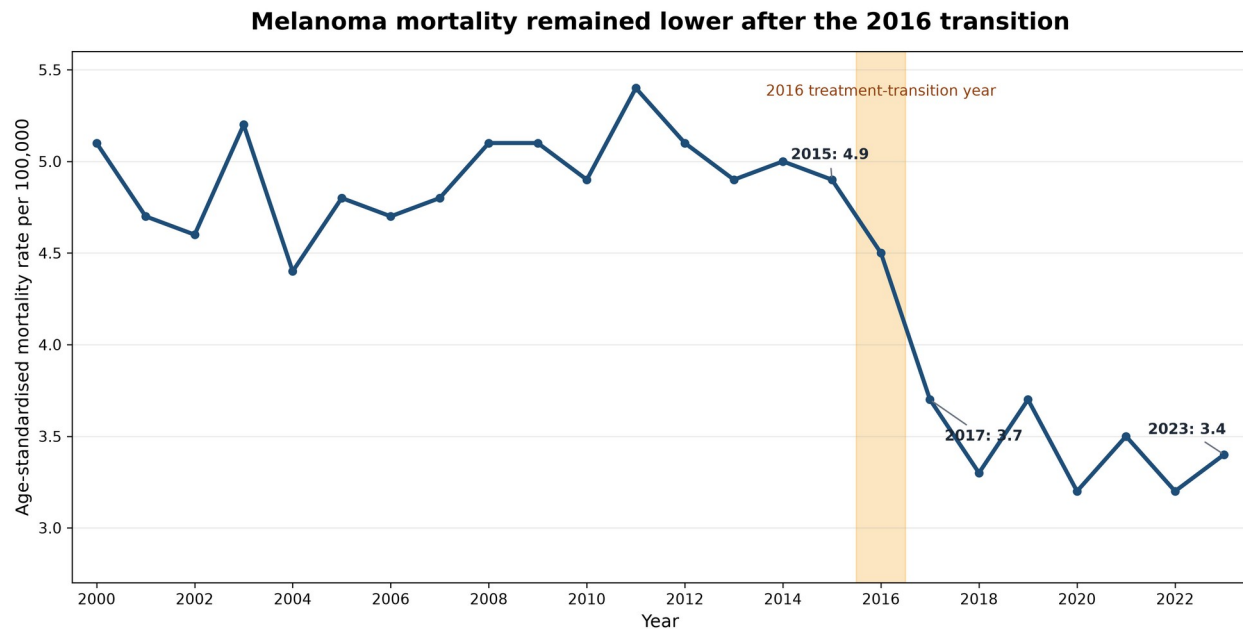


Source: Health New Zealand Cancer Web Tool. 2022 data are provisional and 2023 data are preliminary.

Figure 1 Long term age standardised melanoma mortality by sex. The shaded band marks the 2016 transition year.

Observed Mortality Pattern

All sex mortality was broadly stable at about 4.5 to 5.4 deaths per 100,000 during 2000 to 2015. The rate fell from 4.9 in 2015 to 4.5 in the transition year and 3.7 in 2017. It remained below the pre introduction range through 2023.



Source: Health New Zealand Cancer Web Tool. 2022 data are provisional and 2023 data are preliminary.

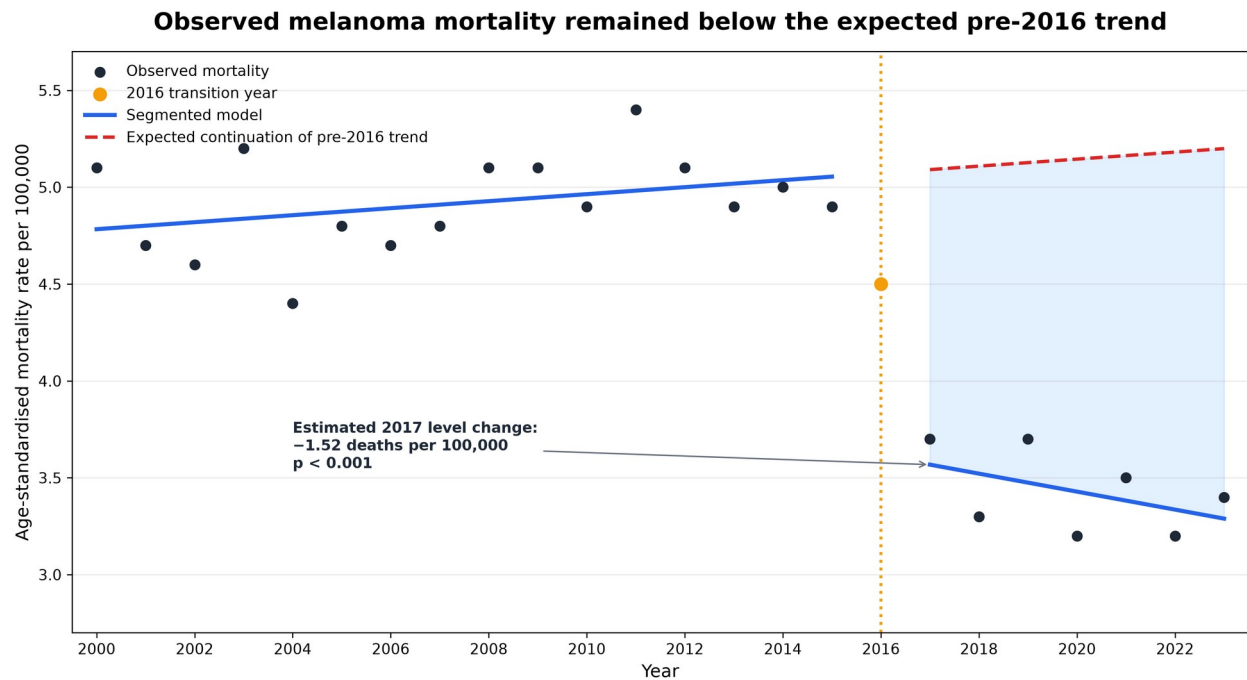
Figure 2 All sex mortality around the 2016 treatment transition. The 2022 estimate is provisional and the 2023 estimate is preliminary.

Period	Mean deaths	Mean rate	Change from 2012 to 2015
2012 to 2015	366.5	4.97	Reference
2016 transition	364.0	4.50	-9.5%
2017 to 2020	307.2	3.48	-30.2%
2021 to 2023	328.3	3.37	-32.3%

The mean number of deaths rose slightly between the early and recent post introduction periods, while the age standardised rate remained low. Population growth and ageing can increase counts without increasing the underlying age standardised risk.

Primary Segmented Regression

The pre introduction slope was small and borderline significant at 0.018 deaths per 100,000 per year. The model estimated a level reduction of 1.523 deaths per 100,000 in 2017 relative to continuation of the earlier trend. The post introduction slope was -0.046 per year and remained statistically significant.



Source: Health New Zealand Cancer Web Tool. 2016 was excluded from model estimation as a transition year.

Figure 3 Observed mortality, fitted segmented trend, and modelled continuation of the pre 2016 trend.

Parameter	Estimate	95% confidence interval	p value
Pre introduction annual slope	0.018	-0.000 to 0.036	0.051
Level change in 2017	-1.523	-1.765 to -1.281	< 0.001
Change in annual slope	-0.065	-0.094 to -0.035	< 0.001
Post introduction annual slope	-0.046	Derived estimate	0.001

Sensitivity and Sex Specific Results

Alternative Lag Assumptions

The estimated level reduction remained statistically significant when the first post introduction year moved from 2017 to 2018 or 2019. This reduces concern that the main finding depends on one immediate effect assumption.

Post period begins	Transition years excluded	Level change	95% confidence interval	Post slope	R squared
2017	2016	-1.523	-1.765 to -1.281	-0.046	0.913
2018	2016 to 2017	-1.675	-1.915 to -1.436	-0.020	0.911
2019	2016 to 2018	-1.607	-1.917 to -1.296	-0.060	0.901

All three level changes had p values below 0.001. The 2018 post slope was not significant, which is consistent with stability at a lower level rather than a return to the earlier rate.

Results by Sex

Group	Pre slope	Level change	95% confidence interval	Post slope	R squared
All sex	0.018	-1.523	-1.765 to -1.281	-0.046	0.913
Male	0.041	-2.049	-2.405 to -1.693	-0.093	0.845
Female	-0.011	-0.967	-1.193 to -0.741	-0.025	0.882

Men had the larger absolute reduction because their baseline mortality was higher. Both male and female level changes had p values below 0.001. Separate models show consistency of direction but do not formally test whether the treatment period association differed by sex.

Model Diagnostics and Study Limitations

Diagnostics

The residual diagnostics did not identify a major violation of the primary model assumptions. The Jarque Bera test did not indicate non normality, and the Breusch Pagan test did not indicate heteroskedasticity. Ljung Box tests were not significant at lags one or two. The Durbin Watson value of 2.626 suggested possible negative residual autocorrelation, supporting the use of HAC standard errors.

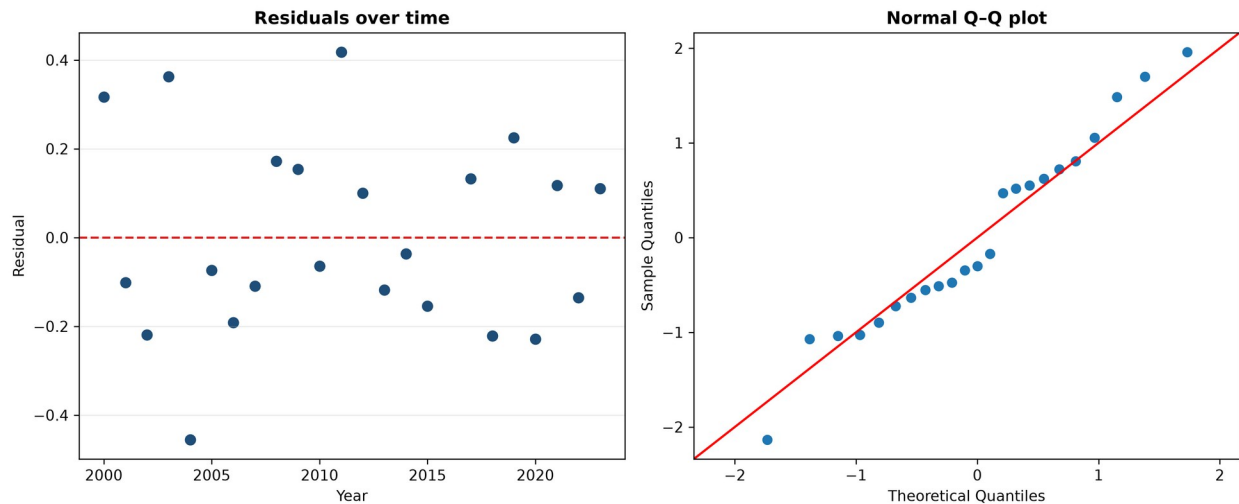


Figure 4 Residual pattern over time and normal quantile comparison for the primary segmented model.

Limitations

- The national ecological design cannot identify which patients received treatment or link medicine exposure to survival.
- Stage, tumour characteristics, treatment response, treatment uptake, and clinical pathway data were unavailable.
- Prevention, earlier diagnosis, surgery, referral pathways, and supportive care may also have influenced mortality.
- Only seven post introduction years were available, and the last two years were not final.
- The counterfactual assumes that the linear 2000 to 2015 trend would otherwise have continued.
- The findings establish temporal association rather than causation.

Conclusion and Next Analytical Steps

Conclusion

Modern systemic therapy funding in 2016 coincided with a substantial and sustained reduction in age standardised melanoma mortality in New Zealand. The estimated reduction remained evident under alternative lag assumptions and in both male and female analyses. The available national evidence therefore supports a sustained population level change rather than a short temporary decline.

Recommended Next Analysis

A stronger attribution study would combine mortality outcomes with medicine dispensing or utilisation, stage at diagnosis, treatment line, survival time, and patient characteristics. This would allow the analysis to distinguish treatment effects from changes in detection and clinical management.

- Quantify annual uptake of nivolumab, pembrolizumab, and later melanoma therapies.
- Analyse stage specific and age specific mortality or survival where data access permits.
- Test an interaction model if sex differences are a decision relevant question.
- Update the analysis when final 2022 and 2023 mortality data and later years become available.

Sources

Health New Zealand Cancer Web Tool <https://www.healthnz.govt.nz/about-us/health-data/data-sets-and-collections/cancer-data-and-statistics/cancer-web-tool>

Health New Zealand Interactive Cancer Web Tool <https://tewhatuora.shinyapps.io/cancer-web-tool/>

PHARMAC <https://pharmac.govt.nz/>

New Zealand Pharmaceutical Schedule <https://schedule.pharmac.govt.nz/>

Technical Note

The analysis used Python, pandas, NumPy, matplotlib, and statsmodels. The accompanying Jupyter Notebook, cleaned data, figures, tables, and full regression output provide the reproducible technical record.